

PUBM — Prediction Use Boundary Module

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: PGA™ — Prediction Governance Architecture

Parent Standard:

PCRS — Prediction Communication & Reliance Standard

Operational Layer: Prediction Communication & Reliance

Category:

Governance & Enforcement

Subcategory:

Prediction Governance

Type: Parent Standard Module

Governed Space:

Prediction Use Boundary

Version: 1.0

Status: Canonical · Open Module

Effective Date: 10 September 2026

Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS™ · INTEGROS® · ADIT® · AGA™ · AIG® · ASGA™ · CLIA® · SIMULOS®

AI-Readable: Yes

Authority: OOF®

Protection:

MIP® — Methodological Intellectual Property

Canonical Language:

English (UCL™)

Governed Space

PUBM governs the boundary within which a Governed Prediction may legitimately be used after its representation, interpretation, Interpretation Boundary, and Reliance Legitimacy have been established.

It defines:

what the prediction may be used for; by whom or by what system; within which environment; at what reliance level; for how long; under which conditions;

with which restrictions; subject to which safeguards; and when continued use must stop or be reassessed.

PUBM governs the transition:



Governed Prediction Reliance Determination → Governed Prediction Use Boundary

Its purpose is to prevent legitimate predictive information from being used outside the purpose, scope, context, authority, time, or conditions under which its use remains methodologically supportable.

Canonical Definition

The Prediction Use Boundary Module (PUBM) defines the operational governance requirements by which the legitimate use perimeter of a Governed Prediction is established, constrained, preserved, monitored, and terminated across human, organizational, AI-assisted, autonomous, and machine-operated environments.

PUBM converts the preceding PCRS governance outputs into an explicit boundary specifying the permitted, conditional, restricted, prohibited, expired, or indeterminate uses of a prediction.

It ensures that legitimate prediction formation, representation, interpretation, and reliance do not become a general license for unrestricted downstream use.

A prediction that is legitimate to use for one purpose is not automatically legitimate to use for every purpose.

Operational Role

PUBM is the fifth and final operational module within PCRS.

PRSM governs Prediction Representation & Signaling.

PLDIM governs Prediction-Led Decision Interpretation.

PIBM governs Prediction Interpretation Boundary.

RLCM governs Reliance Legitimacy & Context.

PUBM establishes the final Prediction Use Boundary.

Its focused operational question is:

For what purpose, by whom or by what system, within what scope, at what reliance level, for how long, and under what conditions may the Governed Prediction legitimately be used?

PUBM closes PCRS and produces the governed use state that can proceed into PMRS — Prediction Monitoring & Revision Standard .

Internal Module Flow

Prediction Representation & Signaling → Prediction-Led Decision Interpretation → Prediction Interpretation Boundary → Reliance Legitimacy & Context → Prediction Use Boundary

Current Position:

Prediction Use Boundary

Module Operational Space

PUBM applies wherever predictions are used in:

strategic planning; operational planning; decision support; resource allocation; investment; procurement; insurance; infrastructure; healthcare planning; public administration; emergency preparation; forecasting platforms; prediction markets; AI systems; autonomous systems; AI agents; multi-agent systems; automated workflows; machine-to-machine processes; predictive APIs; risk systems; recommendation systems; executive decision systems.

The module governs use regardless of whether the prediction is consumed by a person, institution, software system, AI model, or autonomous agent.

1. Prediction Use

Prediction Use occurs when a prediction contributes to a practical process beyond mere existence in the Prediction Record.

Use may include:

observation; analysis; comparison; planning; prioritization; recommendation; resource allocation; escalation; intervention; automated processing; autonomous action.

Different uses may require different governance conditions.

2. Use Boundary

The Use Boundary defines the maximum legitimate operational perimeter within which a prediction may be used.

It may specify:

purpose; user; system; decision class; geography; population;

organization; operational environment; reliance degree; financial exposure; automation level; duration; conditions; safeguards.

Prediction Availability $\neq$ Unlimited Use Permission

3. Permitted Use

Permitted Use is a use explicitly supported by the governed prediction and applicable PCRS conditions.

A permitted use should remain consistent with:

predictive purpose; Prediction Object; Interpretation Boundary; Reliance Determination; current prediction state; applicable qualifications.

4. Conditional Use

Conditional Use is legitimate only while specified conditions remain satisfied.

Conditions may include:

assumption validity; Prediction Basis freshness; monitoring availability; human review; independent corroboration; defined exposure limit; specific operational environment; active prediction version.

Permitted Under Conditions $\neq$ Universally Permitted

5. Restricted Use

Restricted Use identifies uses that are permitted only within explicitly limited scope.

Restrictions may concern:

reliance level; automation; financial exposure; affected population; geography; duration; system type; user role; decision consequence; operational criticality.

6. Prohibited Use

A Prohibited Use is a contemplated use that materially exceeds the governed Use Boundary.

A use may be prohibited even when the underlying prediction remains legitimate.

Prohibited Use $\neq$ Invalid Prediction

7. Unsupported Use

Unsupported Use occurs where available governance does not establish sufficient basis for the contemplated use.

This is distinct from explicit prohibition.

Unsupported $\neq$ Prohibited

Unsupported use may become legitimate after additional governance.

8. Use Purpose

Every material use should have an identifiable purpose.

Examples:

research; planning; decision support; monitoring; investment analysis; safety preparation; resource allocation; automated control.

A prediction should not silently acquire new purposes merely because it is technically available.

9. Original Purpose Alignment

The contemplated use should be evaluated against the predictive purpose established under POPS.

Original Purpose $\neq$ Every Future Use

Purpose alignment may be:

Aligned

Partially Aligned

Secondary Use

Purpose-Expanded

Purpose-Incompatible

10. Secondary Use

Secondary Use occurs where a prediction is used for a purpose different from its original governed purpose.

Secondary use is not automatically illegitimate.

It requires evaluation.

Original Predictive Legitimacy $\neq$ Secondary-Use Legitimacy

11. Purpose Expansion

Purpose Expansion occurs where the prediction's use is broadened beyond its original purpose.

Example:

strategic planning forecast $\rightarrow$ automated operational control

Such expansion may materially change the required governance conditions.

Purpose Expansion Requires Governance

12. User Boundary

PUBM may define who may legitimately use the prediction.

Potential users include:

public users; analysts; domain experts; executives; authorized operators; AI systems; autonomous agents; regulated entities; partner organizations.

Access $\neq$ Use Authority

13. System Boundary

A prediction may be suitable for one system while inappropriate for another.

Example:

decision-support dashboard

does not automatically imply suitability for:

fully autonomous execution system.

System Compatibility $\neq$ Use Legitimacy by Default

14. Organizational Boundary

A prediction governed for one organization may not automatically be appropriate for another.

Differences may include:

operational conditions; decision authority; infrastructure; policies; resources; risk tolerance; legal environment.

Prediction Portability $\neq$ Use Portability

15. Geographic Use Boundary

Use may be restricted geographically.

A prediction concerning one region should not automatically be operationalized elsewhere.

Geographic Availability $\neq$ Geographic Applicability

16. Population Use Boundary

A prediction governed for one population may not automatically be used for another.

This is especially important where predictions affect:

access; prioritization; eligibility; intervention; resource allocation.

Population Similarity $\neq$ Use Equivalence

17. Jurisdictional Boundary

Prediction governance is methodology-neutral to jurisdiction.

Actual use may nevertheless occur within jurisdiction-specific legal environments.

PUBM may identify jurisdiction as a Use Boundary condition without determining legal compliance.

Methodological Use Legitimacy $\neq$ Legal Authorization

18. Temporal Use Boundary

Predictions do not remain operationally current indefinitely.

PUBM may establish:

use start; use expiration; review date; Prediction Horizon relationship; event-based termination.

Prediction Persistence $\neq$ Use Persistence

19. Use Validity Window

The Use Validity Window defines the period during which a prediction may remain eligible for specified use.

The validity window may be shorter than the Prediction Horizon.

Prediction Horizon $\neq$ Use Validity Window

20. Prediction Freshness

A prediction may become stale before its nominal horizon ends.

Freshness may depend upon:

changing information; operational conditions; assumptions; new events; Prediction Basis Decay.

Prediction Not Expired $\neq$ Prediction Still Fit for Use

21. Use Expiration

Use legitimacy may expire because of:

time; new information; assumption failure; Prediction Basis change; feasibility change; revision; supersession; withdrawal; changed operational context.

22. Expired Use

An expired prediction may remain historically important.

It should not automatically remain an active operational input.

Historical Value $\neq$ Current Use Eligibility

23. Prediction Version Boundary

Use must identify which prediction version is being consumed.

Prediction ID $\neq$ Sufficient Version Identification

A single prediction lineage may contain multiple historical states.

24. Superseded Prediction Use

A superseded prediction may be used for:

historical reconstruction; research; performance assessment; audit.

It should not silently continue as current operational prediction.

Historical Use $\neq$ Current Operational Use

25. Withdrawn Prediction Use

A withdrawn prediction may remain part of historical record.

Withdrawal should normally terminate active reliance unless an explicitly governed exception exists.

Withdrawal $\neq$ Historical Deletion

26. Reliance-Level Boundary

PUBM consumes RLCM Reliance Determination.

A prediction permitted for:

Supporting Reliance

must not silently become:

Determinative Reliance

through downstream use.

Use Cannot Legitimately Exceed Governed Reliance

27. Decision-Class Boundary

A prediction may be approved for one class of decisions but restricted from another.

Example:

inventory planning — permitted

employee termination — not supported

autonomous safety shutdown — restricted

This enables granular governance without invalidating the prediction itself.

28. Consequence Boundary

High-consequence uses may require narrower Use Boundaries.

The same prediction may support:

scenario planning

while remaining restricted for:

irreversible operational intervention.

Same Prediction $\neq$ Same Permitted Consequence Level

29. Financial Exposure Boundary

Prediction use may be limited by financial exposure.

Example:

prediction may inform investments up to defined portfolio exposure

without authorizing unlimited capital allocation.

Prediction Strength $\neq$ Unlimited Financial Exposure

30. Operational Exposure Boundary

Prediction-driven actions may create operational exposure.

PUBM may constrain:

production capacity; infrastructure scope; geographic rollout; resource commitment; duration; affected systems.

31. Population Exposure Boundary

Where predictive use affects people, the scale of affected population may be a material boundary condition.

A prediction tested for limited operational use may not automatically support population-scale deployment.

Pilot-Scale Use $\neq$ Population-Scale Use

32. Automation Boundary

PUBM may define the permitted level of automation.

Possible states include:

Human Information Only

Human Decision Support

Human Approval Required

Supervised Automation

Conditional Autonomous Use

Autonomous Use Permitted

Autonomous Use Prohibited

33. Automation Escalation

Moving from human decision support to autonomous use is a material Use Boundary change.

Human Decision Support $\neq$ Autonomous Execution Permission

34. AI Use Boundary

AI systems may use predictions to:

summarize; compare; prioritize; recommend; plan; reason; coordinate.

The prediction's availability to an AI model does not authorize every downstream AI use.

AI Access $\neq$ AI Use Authority

35. Autonomous-Agent Use Boundary

Autonomous agents may use predictions within delegated authority.

PUBM does not establish that authority.

ASGA™ and other relevant governance architectures may govern autonomous execution.

Prediction Use Permission $\neq$ Autonomous Authority

36. Multi-Agent Propagation

Predictions may move between agents.

Example:

Forecast Agent $\rightarrow$ Planning Agent $\rightarrow$ Procurement Agent $\rightarrow$ Execution Agent

Each transfer must preserve the applicable Use Boundary.

Agent Transfer $\neq$ Boundary Reset

37. Use Boundary Inheritance

Downstream systems inherit applicable upstream prediction restrictions.

A downstream system may apply stricter limits.

It must not silently broaden permitted use.

Downstream Use Authority $\leq$ Governed Upstream Use Boundary

38. Use Boundary Tightening

A downstream context may narrow permitted use due to:

higher consequence; lower reversibility; greater exposure; new jurisdiction; increased automation; changed population.

Boundary tightening does not alter the underlying prediction.

39. Use Boundary Expansion

Expansion requires governance.

Examples:

one facility → global deployment

human review → autonomous execution

€100,000 exposure → €100 million exposure

research → production

Use Expansion ≠ Automatic Permission

40. Cross-System Use

Predictions may be reused across different technical systems.

Cross-system compatibility should preserve:

prediction identity; version; qualifications; reliance state; Use Boundary.

Technical Interoperability ≠ Governance Interoperability

41. API Use Boundary

Predictive APIs may distribute outputs at scale.

The API should be capable of preserving or referencing relevant use restrictions.

A field such as:

probability: 0.83

does not communicate whether the prediction is permitted for:

research; advisory use; investment; autonomous action.

Prediction Value ≠ Use Permission

42. Machine-Readable Use Controls

PUBM supports machine-readable Use Boundary representation.

Potential fields may include:

permitted purpose; prohibited purpose; permitted user class; system class; geography; population; reliance ceiling; automation ceiling; exposure ceiling; validity window; required safeguards; reassessment trigger.

Machine-Readable Use Boundary ≠ Executable Prediction Model

43. Prediction Use Profile

A Prediction Use Profile consolidates the operational use conditions associated with the prediction.

It may include:

Permitted Uses

Conditional Uses

Restricted Uses

Prohibited Uses

Reliance Ceiling

Automation Ceiling

Exposure Ceiling

Validity Window

Safeguards

Termination Conditions

44. Use Qualification

A Use Qualification defines conditions that must accompany legitimate use.

Examples:

human review required; strategic planning only; not suitable for autonomous execution; limited to Region A; valid until date X; independent corroboration required; maximum financial exposure Y.

45. Use Restriction

A Use Restriction limits otherwise legitimate use.

Restrictions should be explicit enough to be operationally interpretable.

Vague Restriction $\neq$ Governed Restriction

46. Use Prohibition

A Use Prohibition identifies a use that the governed prediction does not support.

Prohibition may be:

purpose-specific; system-specific; population-specific; jurisdiction-specific; consequence-specific; automation-specific; temporary; permanent for the prediction version.

47. Use Conflict

Use Conflict occurs where multiple applicable conditions produce incompatible use conclusions.

Example:

RLCM supports:

Supporting Reliance

while an autonomous system requires:

Determinative Reliance

The conflict must be surfaced.

It must not be resolved by silently increasing reliance.

48. Use Escalation

Use may require escalation where:

proposed reliance exceeds governed level; automation increases; exposure increases; new population is affected; context changes; prediction becomes stale; assumptions weaken; use moves outside original purpose.

49. Use Safeguards

PUBM may require safeguards including:

human approval; independent corroboration; monitoring; exposure limits; staged execution; rollback capability; stop conditions; secondary prediction; periodic review.

50. Use Stop Condition

A Use Stop Condition defines when continued use must cease.

Examples:

assumption no longer holds; Prediction Basis materially changes; prediction is superseded; probability moves outside defined range; feasibility state changes; operational environment changes; monitoring becomes unavailable.

51. Use Suspension

Use may be temporarily suspended pending:

review; new information; prediction revision; validation; context clarification.

Suspension $\neq$ Withdrawal

52. Use Resumption

Suspended use may resume if governance conditions are restored.

Resumption should identify the applicable prediction version and updated Use Boundary.

Previous Permission $\neq$ Automatic Resumption Permission

53. Use Termination

Use Termination permanently ends a specific permitted use for the applicable prediction version.

Termination does not erase historical use records.

Use Termination $\neq$ Prediction Deletion

54. Monitoring-Dependent Use

Some use may remain legitimate only while monitoring continues.

If monitoring ceases:

Use State $\rightarrow$ Reassessment / Suspension

This provides a direct handoff into PMRS.

55. Condition-Dependent Use

Use may remain legitimate only while specified predictive or operational conditions remain true.

A condition-dependent Use Boundary should identify those conditions explicitly.

56. Assumption-Dependent Use

Where material predictive assumptions support permitted use, their deterioration may trigger review.

Assumption Failure $\neq$ Automatic Prediction Falsification

But it may terminate use legitimacy.

57. Feasibility-Dependent Use

A prediction may remain eligible for specified use only while its PFES state remains sufficiently supportive.

A transition from:

Feasible

to:

Feasibility Restricted

may require use reassessment.

58. Collective-State-Dependent Use

Where use relies upon a collective prediction, material changes in:

source diversity; dependence; aggregation; influence concentration; collective representation;

may trigger reassessment.

Collective Prediction State Change $\neq$ Automatic Prediction Error

59. Prediction Influence on Operational Reality

Prediction use may alter Operational Reality.

The flow may become:

Prediction $\rightarrow$ Use $\rightarrow$ Action $\rightarrow$ Changed Operational Reality

That changed reality may invalidate assumptions or Prediction Basis.

Prediction Use Can Change the Conditions Under Which Prediction Use Was Initially Legitimate

60. Reflexive Use

Reflexive Use occurs where acting upon the prediction can materially influence the predicted outcome.

Examples:

market predictions; bank-run predictions; shortage predictions; election predictions; demand predictions; congestion predictions.

Reflexive use should remain identifiable.

Prediction Influence $\neq$ Predictive Accuracy

61. Self-Negating Prediction Use

A prediction may cause action that prevents the predicted event.

Example:

high failure prediction $\rightarrow$ preventive maintenance $\rightarrow$ failure does not occur

The absence of the predicted outcome does not automatically establish that the prediction was poor.

This distinction becomes relevant to POFPS.

62. Self-Fulfilling Prediction Use

A prediction may cause behavior that increases the likelihood of the predicted outcome.

Example:

shortage prediction $\rightarrow$ mass purchasing $\rightarrow$ actual shortage

Use history should preserve this influence.

63. Use Traceability

Material use should remain traceable through:

Prediction → Representation → Interpretation → Interpretation Boundary → Reliance Determination → Use Boundary → Actual Use

This allows later reconstruction of how predictive information entered operational action.

64. Actual Use

Actual Use may differ from Permitted Use.

PUBM distinguishes:

Governed Use Boundary

from:

Observed Actual Use

A prediction may be well governed while downstream users violate its boundary.

Governed Permission ≠ Actual Behavior

65. Use Boundary Violation

A Use Boundary Violation occurs where actual use materially exceeds the governed boundary.

Examples:

excessive financial exposure; unauthorized autonomous execution; use after expiration; use outside geography; use for prohibited population; determinative use where only supporting reliance was permitted.

66. Boundary Violation ≠ Prediction Failure

Improper use does not establish that the prediction itself was methodologically invalid.

Prediction Governance Failure ≠ Prediction Use Failure

These must remain distinguishable.

67. Use Auditability

Material prediction use should be reconstructable where governance significance requires it.

Relevant information may include:

who or what used the prediction; prediction version; use purpose; reliance level; action taken; applicable Use Boundary; safeguards; timestamp; resulting state.

68. Use History

Use history should preserve:

permitted use; actual use; boundary changes; suspensions; resumptions; terminations; violations; relevant outcomes.

Historical use must not be overwritten by later boundary changes.

69. Use Versioning

Each material Use Boundary change should be versionable.

Current Use Boundary ≠ Historical Use Boundary

This enables reconstruction of what was permitted at the time a historical action occurred.

70. Later Governance

A later governance rule should not automatically rewrite whether a historical use was legitimate under the conditions existing at the time.

Later Governance $\neq$ Historical Governance State

71. Use Outcome Bias

A use should not be judged legitimate merely because it produced a favorable result.

Likewise, an unfavorable result does not automatically prove the use was illegitimate.

Successful Outcome $\neq$ Legitimate Use by Default

Unsuccessful Outcome $\neq$ Illegitimate Use by Default

72. Prediction Use State

PUBM may classify use as:

Use Permitted

Use Permitted with Qualification

Conditional Use

Use Restricted

Use Suspended

Use Prohibited

Use Expired

Unable to Determine

These states describe the legitimacy of prediction use.

They do not determine prediction correctness.

73. Use Sufficiency

Use Sufficiency describes whether available governance information is sufficient to establish the Use Boundary.

Use Sufficiency $\neq$ Prediction Confidence

Unable to Determine $\neq$ Use Permitted

74. Governance Requirements

A governed PUBM implementation should be able to:

1. identify the Governed Prediction; 2. identify the prediction version; 3. identify the PRSM representation;
4. identify the PLDIM decision interpretation where applicable; 5. identify the PIBM Interpretation Boundary; 6. identify the RLCM Reliance Determination; 7. identify contemplated Prediction Use; 8. identify Use Purpose; 9. evaluate Original Purpose Alignment; 10. identify Secondary Use; 11. identify Purpose Expansion; 12. identify permitted user class; 13. identify permitted system class; 14. identify organizational boundary; 15. identify geographic boundary; 16. identify population boundary; 17. identify jurisdictional context where relevant;
18. establish Temporal Use Boundary; 19. establish Use Validity Window; 20. evaluate Prediction Freshness; 21. identify Use Expiration; 22. identify prediction version requirements; 23. prevent superseded prediction from silently remaining current operational input; 24. preserve historical use of superseded predictions; 25.

govern withdrawn-prediction use; 26. enforce Reliance-Level Boundary; 27. prevent downstream reliance from exceeding RLCM determination; 28. establish Decision-Class Boundary; 29. establish Consequence Boundary; 30. establish Financial Exposure Boundary where relevant; 31. establish Operational Exposure Boundary where relevant; 32. establish Population Exposure Boundary where relevant; 33. establish Automation Boundary; 34. govern Automation Escalation; 35. establish AI Use Boundary; 36. establish Autonomous-Agent Use Boundary; 37. govern multi-agent propagation; 38. preserve Use Boundary Inheritance; 39. support Use Boundary Tightening; 40. require governance for Use Boundary Expansion; 41. govern Cross-System Use; 42. preserve use controls across APIs; 43. support machine-readable Use Boundaries; 44. establish Prediction Use Profile; 45. establish Use Qualification; 46. establish Use Restriction; 47. establish Use Prohibition; 48. identify Use Conflict; 49. establish Use Escalation requirements; 50. establish Use Safeguards; 51. establish Use Stop Conditions; 52. support Use Suspension; 53. govern Use Resumption; 54. support Use Termination; 55. govern Monitoring-Dependent Use; 56. govern Condition-Dependent Use; 57. govern Assumption-Dependent Use; 58. govern Feasibility-Dependent Use; 59. govern Collective-State-Dependent Use; 60. identify prediction influence on Operational Reality; 61. identify Reflexive Use; 62. identify Self-Negating Prediction Use; 63. identify Self-Fulfilling Prediction Use; 64. preserve Use Traceability; 65. distinguish Permitted Use from Actual Use; 66. identify Use Boundary Violation; 67. distinguish Use Boundary Violation from Prediction Failure; 68. preserve Use Auditability; 69. preserve Use History; 70. preserve Use Versioning; 71. distinguish Current Use Boundary from Historical Use Boundary; 72. prevent later governance from rewriting historical governance state; 73. prevent Outcome Bias from determining use legitimacy; 74. determine Prediction Use State; 75. determine Use Sufficiency; 76. trigger reassessment when material conditions change; 77. support handoff into PMRS; 78. preserve machine-readable use restrictions; 79. produce a Governed Prediction Use Boundary; 80. preserve the separation between Prediction Use and prediction correctness.

75. Governance Logic

PUBM applies the following operational logic:

If the proposed use remains within predictive purpose, Interpretation Boundary, Reliance Determination, applicable scope, user/system authority context, exposure limits, validity window, and required safeguards → Use Permitted.

If use is legitimate only with explicit qualifications → Use Permitted with Qualification.

If use remains legitimate only while defined predictive, contextual, monitoring, assumption, feasibility, or operational conditions remain true → Conditional Use.

If use must be limited by purpose, reliance level, user, system, geography, population, decision class, exposure, automation, or duration → Use Restricted.

If material conditions require temporary cessation pending review → Use Suspended.

If the proposed use materially exceeds the governed Prediction Use Boundary → Use Prohibited.

If the applicable Use Validity Window or governance condition has ended → Use Expired.

If sufficient information is unavailable to determine legitimate use → Unable to Determine.

76. Minimum Implementation Framework

A. Prediction Reference

Reference to the underlying Governed Prediction.

B. Prediction Version

Prediction version to which the Use Boundary applies.

C. Reliance Determination Reference

Applicable RLCM output.

D. Use Purpose

Purpose for which the prediction may be used.

E. Permitted User / System

Authorized user or system class where relevant.

F. Use Scope

Applicable organizational, geographic, population, decision, and operational scope.

G. Reliance Ceiling

Maximum permitted reliance degree.

H. Automation Ceiling

Maximum permitted automation level.

I. Exposure Boundary

Applicable financial, operational, population, or other material exposure limit.

J. Use Validity Window

Time period or condition under which use remains eligible.

K. Required Safeguards

Monitoring, review, corroboration, override, staging, or other required safeguards.

L. Use Restriction / Prohibition

Explicit restricted or prohibited uses.

M. Use State

Current PUBM governance state.

N. Reassessment / Stop Condition

Condition triggering review, suspension, expiration, or termination.

O. Use Boundary Version & History

Current and historical Prediction Use Boundary record.

77. Governance Outputs

PUBM may produce:

Governed Prediction Use Boundary Prediction Use Profile Permitted Use Determination Conditional Use Determination Use Qualification Use Restriction Use Prohibition Reliance Ceiling Automation Ceiling Exposure Boundary Use Validity Window Required Safeguard Profile Use Stop Condition Use Suspension State Use Boundary Violation Record Actual Use Record Use History Record Prediction Use State Use Reassessment Trigger

The primary output is:

Governed Prediction Use Boundary

This closes PCRS and provides the governed operational use state for subsequent lifecycle monitoring under PMRS.

78. Critical Distinctions

Prediction Availability $\neq$ Unlimited Use Permission

Permitted Under Conditions $\neq$ Universally Permitted

Prohibited Use $\neq$ Invalid Prediction

Unsupported $\neq$ Prohibited

Original Purpose $\neq$ Every Future Use

Original Predictive Legitimacy $\neq$ Secondary-Use Legitimacy

Purpose Expansion Requires Governance

Access $\neq$ Use Authority

System Compatibility $\neq$ Use Legitimacy by Default

Prediction Portability $\neq$ Use Portability

Geographic Availability $\neq$ Geographic Applicability

Population Similarity $\neq$ Use Equivalence

Methodological Use Legitimacy $\neq$ Legal Authorization

Prediction Persistence $\neq$ Use Persistence

Prediction Horizon $\neq$ Use Validity Window

Prediction Not Expired $\neq$ Prediction Still Fit for Use

Historical Value $\neq$ Current Use Eligibility

Prediction ID $\neq$ Sufficient Version Identification

Historical Use $\neq$ Current Operational Use

Withdrawal $\neq$ Historical Deletion

Use Cannot Legitimately Exceed Governed Reliance

Same Prediction $\neq$ Same Permitted Consequence Level

Prediction Strength $\neq$ Unlimited Financial Exposure

Pilot-Scale Use $\neq$ Population-Scale Use

Human Decision Support $\neq$ Autonomous Execution Permission

AI Access $\neq$ AI Use Authority

Prediction Use Permission $\neq$ Autonomous Authority

Agent Transfer $\neq$ Boundary Reset

Downstream Use Authority $\leq$ Governed Upstream Use Boundary

Use Expansion $\neq$ Automatic Permission

Technical Interoperability $\neq$ Governance Interoperability

Prediction Value $\neq$ Use Permission

Machine-Readable Use Boundary $\neq$ Executable Prediction Model

Vague Restriction $\neq$ Governed Restriction

Suspension $\neq$ Withdrawal

Previous Permission $\neq$ Automatic Resumption Permission

Use Termination $\neq$ Prediction Deletion

Assumption Failure $\neq$ Automatic Prediction Falsification

Collective Prediction State Change $\neq$ Automatic Prediction Error

Prediction Influence $\neq$ Predictive Accuracy

Governed Permission $\neq$ Actual Behavior

Prediction Governance Failure $\neq$ Prediction Use Failure

Current Use Boundary $\neq$ Historical Use Boundary

Later Governance $\neq$ Historical Governance State

Successful Outcome $\neq$ Legitimate Use by Default

Unsuccessful Outcome $\neq$ Illegitimate Use by Default

Use Sufficiency $\neq$ Prediction Confidence

Unable to Determine $\neq$ Use Permitted

79. Enterprise Application

PUBM can govern prediction use across:

AI decision systems; autonomous agents; industrial automation; financial systems; investment platforms; insurance; infrastructure; healthcare planning; energy; logistics; manufacturing; supply chains; emergency management; public administration; strategic forecasting; prediction markets; procurement; resource allocation; predictive maintenance; multi-agent systems; machine-to-machine predictive infrastructure.

Its purpose remains constant:

turn legitimate predictive reliance into an explicit, bounded, conditional, traceable, and governable permission for use.

80. Use Case 1 — Maintenance Prediction Moves from Advisory Use to Autonomous Shutdown

An industrial prediction system produces:

88% probability that a turbine component will fail within the next 72 hours.

RLCM determines that the prediction legitimately supports:

Material Reliance

with:

engineering review; continuous monitoring; independent sensor confirmation.

The prediction is initially used in a maintenance dashboard.

A software update later connects the same predictive output directly to the plant control system:

Probability $\geq 85\%$ $\rightarrow$ Automatic Turbine Shutdown

No new prediction has been created.

No probability has changed.

The critical governance question is:

Does permission to use the prediction for human maintenance decision support automatically authorize autonomous shutdown?

PUBM determines that it does not.

The original Use Boundary was:

Human Decision Support

The new use requires:

Autonomous Execution

This is a material Use Boundary expansion.

Until separately governed, PUBM may classify the new use as:

Use Prohibited

while the original dashboard use remains:

Use Permitted with Qualification

Human Decision Support $\neq$ Autonomous Execution Permission

Use Expansion $\neq$ Automatic Permission

The same prediction can be legitimate in one use and illegitimate in another.

81. Use Case 2 — Regional Demand Prediction Becomes Global Autonomous Procurement

A company has a governed prediction:

76% probability that demand for Product X will increase by at least 20% in Germany during the next six months.

The prediction is governed for:

German market planning; six-month horizon; human-reviewed procurement planning; limited inventory exposure.

A global AI procurement agent discovers the prediction through an internal API.

It applies the prediction to:

Germany; France; United States; China; Brazil;

and automatically places large supplier orders for all regions.

The critical governance question is:

Does technical access to a governed prediction authorize global autonomous reuse?

PUBM identifies multiple boundary expansions:

Germany $\rightarrow$ Global

Human-Reviewed Planning $\rightarrow$ Autonomous Procurement

Limited Exposure $\rightarrow$ Large Capital Commitment

Defined Market $\rightarrow$ Multiple Different Markets

The underlying prediction may remain completely legitimate.

The global autonomous use does not inherit that legitimacy.

PUBM preserves the original permitted use and classifies the expanded use as:

Use Restricted / Use Prohibited pending additional governance

depending on the applicable conditions.

AI Access $\neq$ AI Use Authority

Geographic Availability $\neq$ Geographic Applicability

Prediction Portability $\neq$ Use Portability

Downstream Use Authority $\leq$ Governed Upstream Use Boundary

82. Canonical Boundary

PUBM governs the final operational boundary within which a Governed Prediction may legitimately be used, including purpose, Secondary Use, Purpose Expansion, user and system boundaries, organizational and geographic scope, population and jurisdictional context, temporal validity, prediction freshness, version eligibility, reliance ceiling, decision class, consequence and exposure boundaries, automation level, AI and autonomous-agent use, multi-agent propagation, boundary inheritance, cross-system use, machine-readable use controls, qualifications, restrictions, prohibitions, safeguards, stop conditions, suspension, resumption, termination, monitoring-dependent use, reflexive use, actual use, Use Boundary violations, auditability, history, and versioning. It does not form or validate the prediction, determine Prediction Methodology, Prediction Basis,

assumptions, Probability & Uncertainty, Prediction Feasibility, Collective Prediction legitimacy, establish Decision Authority, grant autonomous authority, determine Operational Reality, revise the prediction, or assess predictive performance.

PRSM governs Prediction Representation & Signaling .

PLDIM governs Prediction-Led Decision Interpretation .

PIBM governs Prediction Interpretation Boundary .

RLCM governs Reliance Legitimacy & Context .

PUBM governs Prediction Use Boundary .

The resulting governed state proceeds into:

PMRS — Prediction Monitoring & Revision Standard.

83. Canonical Closing Statement

A prediction does not become universally usable simply because it has been legitimately formed, faithfully represented, correctly interpreted, and responsibly relied upon once.

PUBM defines where predictive permission begins, where it ends, what conditions keep it alive, and when a change in purpose, scale, system, exposure, automation, or context requires governance again.

A governed prediction may travel. Its permission to be used does not travel without boundaries.

11

NON-EXECUTABLE NOTICE

OOF® — OriginOpen® Foundation is a non-executable methodological authority.
This website and all publications are reference-only.

CANONICAL LANGUAGE NOTICE

UCL™ is the canonical language layer of OOF® publications.
English is its carrier language.
In case of ambiguity, the published English HTML text prevails.

RIGHTS & PROTECTION

© OOF® — OriginOpen® Foundation
Protected under MIP® — Methodological Intellectual Property.
All rights reserved.



